

Grade 8
Unit 3
Lesson 4 Practice

Name Answer Key
Date _____
Period _____

1. Consider the value 4.8×10^{-6} . Complete the statements. Then express this value in standard form.

10^{-6} relates to the millionths place. When rewriting 4.8×10^{-6} in standard form, the 4 will be in the millionths place, followed by the 8. The places before the 4 will each hold the digit, 0.

2. The diameter of an atom in gold is about 1.44×10^{-10} meters. Express this value as a decimal in standard form.

0.000000000144

3. The average mass of one ant is about 2.9×10^{-6} kilograms. Express this value in standard form.

0.0000029

Express each number in scientific notation.

4. The diameter of a human hair is about 67 hundred-thousandths of an inch.

6.7×10^{-4}

5. 0.0000000049

4.9×10^{-9}

Express each number in standard form.

6. A cell's diameter 0.0174×10^{-4} 

0.00000174

7. 9×10^{-6}

0.000009

8. Complete the table.

Scientific Notation	Standard Form
2.6×10^{-8}	0.000000026
5.2×10^{-3}	0.0052

9. Complete the statements using the table in question 8.

○ 5.2 is 2 times

○ smaller
☒ larger

than 2.6.

○ 10^{-3} is 10^5 times

○ smaller
☒ larger

than 10^{-8} .

○ The value 5.2×10^{-3} is 2×10^5 times or 200,000

○ smaller
☒ larger

than 0.000000026.

10. Match each value on the left with its expression in scientific notation on the right by drawing lines to connect the equivalent expressions.

